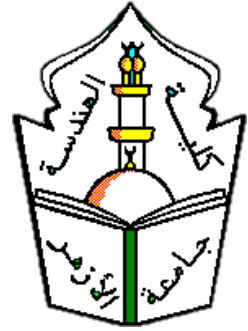




Al-Azhar University
Faculty of Engineering
Systems and Computers Engineering
Department



Smart AI Library - Intelligent Book Recommendation System (Learnova)

Prepared by:

Reham Ehab El-Sayed Ahmed Barakat

Mariam Mohamed Abdullah Mohamed

Norhan Abdelrahman Fathy Salam

Hend Mohamed Ahmed Abdel Motaleb

Wafaa Yousry Ahmed Kohia

Salma Ibrahim Ali

Supervised by:

Dr. Sayed Noah

July 2026

Acknowledgments

In the name of Allah, the Most Gracious, the Most Merciful.

All praise and thanks are due to Allah, the Lord of the worlds, for His countless blessings and for granting us the strength and knowledge to complete this project. Without His guidance and mercy, none of this would have been possible.

We express our deepest gratitude to our beloved families for their unwavering support, prayers, and encouragement throughout our academic journey. Their love and faith have been a constant source of motivation and strength.

We are profoundly grateful to our esteemed supervisor, **Dr. Sayed Noah**, for his invaluable guidance, patience, and insightful feedback. His expertise and dedication were instrumental in shaping the quality and direction of this work.

We also extend our sincere thanks to our teachers and mentors who have imparted wisdom and knowledge throughout our studies. Their efforts have contributed significantly to our personal and academic growth.

May Allah reward everyone who supported us in this endeavor and bless them abundantly.

Alhamdulillah.

Abstract

In the contemporary digital learning landscape, the challenge of delivering personalized, accessible, and time-efficient education has become increasingly critical. Modern learners face a fundamental paradox: unprecedented access to knowledge paired with diminishing time to engage with it. This project presents **Learnova** — an AI-powered Audiobook and E-Learning platform designed to bridge the gap between fragmented audiobook services and structured educational environments through intelligent automation and audio-first delivery.

The platform enables users to transform PDF books and educational materials into immersive, high-quality audio experiences via a state-of-the-art Text-to-Speech (TTS) pipeline, delivering a Spotify-inspired listening interface with persistent Mini Player support, adjustable playback speed, and chapter-level navigation. Beyond audio delivery, Learnova integrates artificial intelligence capabilities including a context-aware conversational chatbot, AI-generated chapter summarization, and a personalized recommendation engine — ensuring each user receives an adaptive and tailored learning journey.

The system is built on a modern cross-platform mobile architecture using **Flutter** and **Dart** for the frontend with Bloc state management, backed by a **PHP Laravel** RESTful API and **MySQL** database. AI features are powered by large language model APIs including OpenAI and Gemini, integrated through a LangChain-based RAG pipeline. Real-time capabilities are supported via Socket.io and Laravel Echo, while secure authentication is handled through JWT and Laravel Sanctum.

This project demonstrates the practical feasibility of unifying audio-first consumption with structured e-learning progression, establishing Learnova as a foundational model for next-generation educational platforms that prioritize accessibility, personalization, and learner autonomy.

Contents

Acknowledgments	i
Abstract	ii
1 Introduction	1
1.1 Overview	1
1.2 Background	1
1.3 Problem Statement	1
1.4 Objectives	1

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Overview

Your text here...

1.2 Background

Your text here...

1.3 Problem Statement

Your text here...

1.4 Objectives

Your text here...

Bibliography
